

1. Let G be a group. We define $*$: $(G \times G) \times G \rightarrow G$ by $(g, h) * x = gxh^{-1}$.

- (a) Prove that $*$ is a group action by $G \times G$ on G .
- (b) Prove that all elements of G belong to the same orbit.
- (c) Find $S(1)$.

(a) Let $(g_1, h_1), (g_2, h_2) \in G \times G$ and $x \in G$, then

$$\begin{aligned}(g_1, h_1) * ((g_2, h_2) * x) &= (g_1, h_1) * (g_2 x h_2^{-1}) \\ &= g_1 g_2 x h_2^{-1} h_1^{-1} \\ &= (g_1 g_2) x (h_1 h_2)^{-1} \\ &= (g_1 g_2, h_1 h_2) * x \\ &= ((g_1, h_1)(g_2, h_2)) * x.\end{aligned}$$

Moreover, $(1, 1) * x = 1x1^{-1} = x$.

- (b) It suffices to show that for arbitrary elements $x, y \in G$, we have $x \in (G \times G) * y$. That is, there exists some $(g, h) \in G \times G$ such that $x = (g, h) * y$. Taking $(g, h) = (x, y)$, we have $(x, y) * y = xy y^{-1} = x$, as desired.
- (c) $S(1) = \{(g, g) : g \in G\}$.

[3 + 5 + 2 = 10 marks]

2. (a) Prove the following lemma:

Lemma. *If a finite nontrivial group has no nontrivial proper subgroups, then it has prime order.*

- (b) Let G be a finite group with exactly three subgroups. Prove that G has order p^2 for some prime p .

HINT: Examine the nontrivial proper subgroup of G . What is its order? Is it normal? What would the order of its factor group be? Make use of the lemma.

- (a) Since G is nontrivial, there is some prime p that divides the order of G . By Cauchy's Theorem, G then has a subgroup of order p . If G has no nontrivial proper subgroups, we then require this subgroup to be G . That is, G has order p .
- (b) Since G has exactly three subgroups, they must be $\{1\}$, G and H (where H is nontrivial and proper). Notice that H has no nontrivial proper subgroups since then that would be another subgroup of G . Thus H has order p for some prime p . For any $g \in G$, we have that gHg^{-1} is another subgroup of G with order p . However, the only subgroup of G with such order must be H . Thus $gHg^{-1} = H$ for all $g \in G$. It follows that $H \triangleleft G$. Now consider G/H . By the Correspondence Theorem, the only subgroups are G/H and H/H . So G/H has no nontrivial proper subgroups and so it has prime order, say q . By Lagrange's Theorem, we then have $|G| = |H||G/H| = pq$. By Cauchy's Theorem, G then has a subgroup of order q . This is only possible if $p = q$ (and thus the subgroup of order $q = p$ is actually H). Thus $|G| = p^2$.

[5 + 15 = 20 marks]

3. (a) Let G be a p -group and $*$ be a group action by G on a finite set X .

(i) State the definition of X_f .

(ii) Write a statement that relates $|X|$ to $|X_f|$, as seen in class.

(i) $X_f = \{x \in X : g * x = x \text{ for all } g \in G\}$

(ii) p divides $|X| - |X_f|$

(b) Let G be a p -group and H a proper subgroup of G .

(i) State the definition of $N_G(H)$.

(ii) Prove that $N_G(H) \neq H$. HINT: Let H act on the set of left cosets of H in G through left multiplication (you do not need to prove this is a group action).

(i) $N_G(H) = \{g \in G : gHg^{-1} = H \text{ for all } g \in G\}$

(ii) Let X be the set of left cosets of H in G . Let H act on X through left multiplication: $h * gH = hgH$. Notice that $H \in X_f$ since $h * H = H$ for all $h \in H$ and so $|X_f| \geq 1$. We know that (since H is a p -group action on X) p divides $|X| - |X_f|$. However, $|X| = |G : H| = |G|/|H|$. Since G is a p -group and H a proper subgroup, we have that p divides $|X|$. Thus $|X_f| > 1$.

We have

$$\begin{aligned} gH \in X_f &\iff hgH = gH \text{ for all } h \in H \\ &\iff g^{-1}hg \in H \text{ for all } h \in H \\ &\iff g^{-1}Hg = H \\ &\iff g \in N_G(H) \end{aligned}$$

Since $|X_f| > 1$, there is some $gH \in X_f$ with $gH \neq H$. That is, $g \notin H$. So there is some $g \in G$ such that $g \in N_G(H)$ and $g \notin H$. Thus $N_G(H) \neq H$.

[2 + 2 + 2 + 14 = 20 marks]

4. (a) State the Third Sylow Theorem.

(b) Let G be a group of order $217 = 7 \times 31$. Prove or disprove: G is cyclic.

HINT: Determine if there are any elements of order 217.

(a) **Third Sylow Theorem.** *Let G be a group of order $p^n m$ where $n \geq 1$ and p is prime with $p \nmid m$. Let n_p denote the number of Sylow p -subgroups of G . Then $n_p \equiv 1 \pmod{p}$ and $n_p \mid m$.*

(b) By the Third Sylow Theorem, $n_7 \equiv 1 \pmod{7}$ and $n_7 \mid 31$. Since 31 is prime, we must have $n_7 = 1$. Similarly, $n_{31} \equiv 1 \pmod{31}$ and $n_{31} \mid 7$ and thus $n_{31} = 1$. This means that there is a unique Sylow 7-subgroup, consisting of 6 elements of order 7. Furthermore, there is a unique Sylow 31-subgroup, consisting of 30 elements of order 31. Any element of G with order 7 or 31 must fall into these Sylow p -subgroups, otherwise they would generate a new Sylow p -subgroup. The identity has order 1. The remaining $217 - 30 - 6 - 1 = 180$ elements of G must have an order that divides the order of G . Thus there are 180 elements of G with order 217, meaning G is cyclic.

[3 + 10 = 13 marks]